Decoding Currency Dynamics

AI-Driven Multi-Step Forecasting of Foreign Exchange Rates

Mid-Term Technical Summary — a companion to the presentation

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1. Project Overview & Presentation at a Glance

This dissertation builds an AI-driven framework for **multi-step forecasting of the XAU/USD (gold) exchange rate**. The core is a **Hybrid CNN-LSTM-Transformer** that fuses three real data streams — price/technical indicators, macroeconomic variables, and FinBERT news sentiment — and forecasts the next 10 daily steps together with a calibrated uncertainty band. The work is positioned against classical econometric baselines (ARIMA, GARCH) and evaluated honestly under a leakage-free, regime-aware protocol.

The accompanying 23-slide deck moves from the problem and objectives, through the data pipeline (price, news, macro), FinBERT sentiment scoring, technical indicators and feature engineering, into the layer-by-layer model architecture and its integration, the system design, tooling, results and honest limitations, and finally the future plan, timeline and conclusion. This document expands the technical core of that deck.

2. Model Architecture — the Hybrid CNN-LSTM-Transformer

The network is a **dual-tower** design (≈4.39M parameters). A quantitative tower and a text tower each encode a 60-bar lookback window, are fused by cross-attention, then passed through a Transformer, parallel recurrent branches, and regime-conditioned probabilistic output heads. Tensor shapes are shown at each hand-off (B = batch).

2.1 Tower A — Quantitative pathway

- **Input (B, 60, 18):** 12 technical + 6 macroeconomic features per bar, projected to width 64.
- **Dilated causal CNN → (B, 60, 128):** stacked 1-D convolutions with dilations 1/2/4 and left-padding (WaveNet/TCN style). It is *causal* (no look-ahead) and uses **no pooling**, so the full 60-bar temporal resolution is preserved while an exponentially growing receptive field captures short, localized structural breaks that a purely recurrent model smooths over.
- A learned **volatility-regime embedding** is added, so downstream layers see the current regime.

2.2 Tower B — News/sentiment pathway

- **Input (B, 60, 13):** 13 FinBERT sentiment features (rolling statistics, a diffusion breadth index, and the one-hot buy/sell/hold/none signal), encoded by a **GRU → (B, 60, 128)**.

2.3 Cross-attention fusion

The two towers are fused by **multi-head cross-attention** with the *price* stream as Query and the *news* stream as Key/Value, so each price position can read the relevant news context. Because news is absent on most bars, a per-timestep **text-presence gate** (a sigmoid driven by the raw sentiment) scales the news contribution on a residual path — on news-less bars the gate closes and the model falls back to the quantitative signal. Output: (B, 60, 128).

2.4 Transformer, recurrent blend, and pooling

- **Transformer encoder → (B, 60, 256):** multi-head self-attention with positional encoding and a **causal mask**. Placing global attention *before* the recurrent stage lets every position attend to every other, capturing long-range, multi-scale dependencies without the recency bias of a pure RNN.

- **Bi-LSTM ‖ Bi-GRU → (B, 256):** two parallel bidirectional recurrent branches blended by a learned gate form the temporal backbone; an attention pool condenses the sequence to a single context vector.

2.5 Regime-aware probabilistic output & the fused XGBoost expert

- A **volatility-regime detector** routes the pooled context to **regime-conditioned decoder heads**; each head emits a mean μ and a log-variance log σ^2 for every one of the 10 horizons (see §3.2).
- A frozen **XGBoost** forecast is blended in as an internal expert through a regime-driven, per-horizon **trust gate**: forecast = trust ⊙ xgb_pred + (1 − trust) ⊙ deep_forecast — the tabular expert's strengths complement the deep pathway, and the gate decides how much to trust each per regime.

3. Key Training Techniques

3.1 Two-Stage Freeze-and-Tune training

News exists for only a fraction of the 26-year history, so training the text tower end-to-end over all bars would project ~17 news-less years of zeroed sentiment onto the fusion and corrupt it. The remedy is a two-stage schedule:

Stage 1 (quant backbone): the quantitative tower, fusion, Transformer, recurrent branches and heads are trained *text-free over the full history*, learning price/macro dynamics on maximum data.

Stage 2 (freeze & tune): the quant tower (~4.15M parameters) is **frozen** and only the text tower, cross-attention fusion and decoder heads (~0.24M parameters) are fine-tuned on the *news-dense* recent subset at 0.1× learning rate. Sentiment refines a well-formed forecast instead of destabilising the backbone.

3.2 Gaussian NLL loss (probabilistic / GARCH-emulating heads)

Rather than a single point value, each decoder head predicts a **distribution** — a mean μ and a log-variance log σ^2 per horizon — trained under the **Gaussian negative log-likelihood**:

$$L = \tfrac{1}{2} \cdot [\, \log \sigma^2 + (y - \mu)^2 / \sigma^2 \,]$$

The variance term lets the network learn *how uncertain* each forecast is, effectively emulating GARCH's conditional-variance modelling *inside* the neural net. The predicted σ becomes a volatility band, and a **conviction** statistic (t-ratio $|\mu| / \sigma$) drives the calibrated-abstention rule and the costed backtest — high-uncertainty forecasts are automatically down-weighted.

3.3 Modality masking (sentiment dropout)

During training the entire sentiment stream is randomly zeroed with probability **p = 0.4**. Because ~82% of real bars carry no news, the model must forecast well *without* sentiment; modality masking prevents it from over-relying on the text tower and teaches a graceful fallback to the quantitative signal. It works hand in hand with the presence gate (§2.3), which realises the same behaviour at inference time.

3.4 Deep supervision

An auxiliary forecast head is attached to a deeper intermediate expert and supervised with its own loss (weight 0.5) in addition to the final-layer loss. This injects gradient signal directly into the middle of the deep CNN→Transformer→recurrent stack, improving gradient flow, mitigating vanishing gradients, and stabilising training of an otherwise long forward path.

Supporting choices: a **Huber** robust regression term (resistant to fat-tailed FX outliers), a small **directional loss** that rewards getting the sign right, and stationary feature transforms so no layer sees absolute price levels.

4. Why Classical Baselines Fall Short — and How the Hybrid Responds

ARIMA and GARCH remain the canonical econometric benchmarks, but their modelling assumptions are routinely violated by currency data. The table maps each limitation to the corresponding mechanism in the Hybrid.

Model	Core limitation	How the Hybrid handles it
ARIMA	Linear & assumes stationarity — FX has a unit root (ADF confirms non-stationarity) and non-linear dynamics ARIMA cannot represent.	Non-linear CNN + LSTM/GRU + Transformer learn multi-scale patterns; stationary feature transforms and regime routing adapt to changing dynamics.
ARIMA	Univariate — uses only the price's own past; ignores macro drivers and news.	Multi-modal fusion of technical + macro + FinBERT sentiment streams, so exogenous drivers inform every forecast.
ARIMA	Multi-step forecasts decay quickly (recursive error propagation, fast mean reversion).	Direct multi-horizon heads emit all 10 steps jointly (no recursion); probabilistic heads quantify per-horizon uncertainty.
GARCH	Strong on variance, weak on the mean — its directional edge is essentially a linear AR(1) momentum drift.	Deep quant tower + fused XGBoost expert capture non-linear drift and interactions; the honest gap to GARCH is narrowed and paired with a conviction filter.
GARCH	Linear-Gaussian, fixed parametric form; no exogenous features, no learned representations, no news.	Learns representations end-to-end and ingests news/macro; cross-attention lets sentiment modulate the price forecast.
Both	No regime adaptation — one fixed model across calm and turbulent markets.	Explicit volatility-regime detector conditions the output heads; evaluation is regime-segmented.
Both	Point forecasts only — no native, per-horizon uncertainty for decision-making.	Gaussian-NLL heads output a (μ, σ^2) band per horizon; conviction = $\lvert\mu\rvert/\sigma$ powers abstention and a costed backtest.

An honest note on GARCH. On this strongly trending, volatility-clustered daily gold series, GARCH's momentum-drift is a genuinely strong *directional* baseline that the Hybrid does not overtake on raw accuracy. The Hybrid's advantage is breadth and decision quality: multi-modal inputs, learned non-linear structure, native uncertainty, regime awareness, and interpretability — plus a conviction/consensus layer that turns forecasts into a positive costed strategy.

5. Results & Honest Findings

Model	Directional Acc. ↑	MAE ↓	Role
GARCH (AR(1)-GARCH(1,1))	0.575	0.0196	Econometric baseline
ARIMA (walk-forward)	0.485	0.0199	Econometric baseline
Hybrid CNN-LSTM-Transformer	**0.534**	**0.0221**	Proposed multi-modal model

- **Daily XAU/USD, 962 test windows, 3 seeds** — every model scored on identical, full-resolution walk-forward origins.
- GARCH leads unfiltered directional accuracy; the Hybrid narrows the gap (0.500 → 0.534 after the sentiment refinement) with near-zero seed variance, and achieves the best MAE among the deep configurations.
- The Hybrid's genuine edge is its **conviction machinery** — the (μ, σ^2) uncertainty, calibrated abstention and a consensus filter deliver a positive costed backtest — rather than raw accuracy.
- **Binding constraint:** news covers only ~18% of test bars (free GDELT tier). Every part of the sentiment pipeline is now correct and industry-aligned, but denser coverage (a keyed news API) is the main remaining lever.

- **Integrity:** >0.85 directional accuracy is not attainable on this task without look-ahead leakage — it is documented as such, not chased.

6. Conclusion

The project delivers a complete, reproducible, multi-modal FX forecaster that unifies CNN spatial extraction, recurrent temporal modelling, Transformer global attention and FinBERT sentiment fusion into one regime-aware, probabilistic architecture — closing the gaps identified in the literature survey (single-purpose pipelines, under-studied multi-step forecasting, and blended-regime evaluation). The freeze-and-tune schedule, Gaussian-NLL heads, modality masking and deep supervision are the techniques that make training on sparse, heterogeneous data stable and honest. Classical baselines remain hard to beat on raw direction, but the Hybrid's multi-modal reach, native uncertainty and conviction machinery make it the more decision-useful and interpretable model — with a clear path (denser sentiment, XAI, ablations, an RL execution layer) to final submission.

Appendix A. Key Training Techniques in Plain English

The same four techniques from Section 3, explained without the jargon — what each one is, why it helps in general, and what it specifically buys this project.

1. Two-Stage Freeze-and-Tune

“ Like teaching a student: first build a rock-solid maths foundation from a thick textbook, then lock that in so it can’t be forgotten — and only afterwards teach the specialised news-reading elective from newer material. ”

In plain terms: Train in two phases — first the price/maths half of the model on all 26 years of data (news ignored), then **freeze** it so it can’t change and train *only* the news-reading half, gently, on the recent years where news actually exists.

Why it helps: When you have lots of data for one thing and very little for another, training everything at once lets the scarce, noisy signal corrupt what you already learned well. Splitting the training protects the strong part.

In our project: News exists on only ~18% of days, so the ~17 news-less years would degrade a jointly-trained model. Freeze-and-tune keeps the price model rock-solid and lets the sparse news gently refine it — a key reason our model trains stably.

2. Gaussian NLL Loss (predicting with confidence)

“ A forecaster who says “70% chance of rain” is far more useful than one who just says “rain.” ”

In plain terms: Instead of a single number, the model predicts a value *and* how sure it is (a confidence range). The loss formula is simply the grading rule that rewards being accurate *and* honest about confidence — it punishes being **confidently wrong** the hardest.

Why it helps: Real decisions need to know the risk, not just the prediction. A single number hides uncertainty; a range lets you act only when the model is genuinely confident.

In our project: GARCH is famous for modelling market **volatility**; we bake that same ability directly into the network — it learns the expected move *and* the expected volatility. That gives a **conviction score** that powers our trade filter (only trade when confident), which the backtest shows turns a profit.

3. Modality Masking (don’t depend on news)

“ Like training a driver to cope when the GPS randomly cuts out, so they rely on the road and treat GPS as a bonus, not a crutch. ”

In plain terms: During training we randomly **hide the news input 40% of the time**, forcing the model to predict well even with the news blacked out.

Why it helps: If an input is often missing or unreliable, a model that leans on it will break when it disappears. Randomly hiding it during training builds robustness and prevents over-reliance.

In our project: News is absent on ~82% of days, so an over-reliant model would fail most days. Masking teaches the model to forecast from price + macro alone and use news as a bonus. It pairs with the **presence gate** — an automatic dimmer that turns the news channel down on no-news days.

4. Deep Supervision (feedback for the middle)

“ On a long assembly line, add a quality checkpoint in the middle instead of only inspecting at the very end. ”

In plain terms: Our network is very deep (many stacked layers). We attach an extra **mini-exit partway through** that also makes a prediction and gets graded, so the middle of the network receives direct feedback — not just the end.

Why it helps: In deep networks the learning signal weakens before it reaches the middle layers (the “vanishing gradient” problem). A mid-network checkpoint injects fresh signal so those layers actually learn and training stays stable.

In our project: Our forward path is long (CNN → cross-attention → Transformer → LSTM/GRU → heads). Deep supervision keeps the middle layers learning and helps the model settle on a consistent solution — seen as the very low seed-to-seed variance across our training runs.